



BUSINESS BLUEPRINT

AOP AUTOMATION FOR VMD EOEG PLANT

Honeywell

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Business Blueprint

AOP Automation for VMD EOEG Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)



Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	12-Nov-2024	HON	Issued for Review
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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for VMD EOEG plant for VMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various EOEG plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for EOEG technology.

1.2 Reference Documents

Document Name	Revision	Title	Document
VMD EOEG DCT	-	Digital AOP VMD EOEG DCT	 DCT VMD MEG.xlsx
UAT Observation list	-	Digital AOP EOEG UAT Observation list	 Digital AOP MEG UAT Observations.xlsx
EOE effective Operating Hours calculation	-	RIL AOP - EOEG Maintenance Screen - Slowdown Logic_R&T Validated	 RIL AOP - MEG Maintenance Screen
Slowdown Norms Methodology	-	Digital AOP - EOEG - Normal_slowdown_shutdown_Sample Norms Calculation	 Digital AOP - EOEG - Normal_slowdown_sf

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Overall AOP Consumption

Steps to generate the Production and Overall AOP Consumption are captured in the below flow diagram.

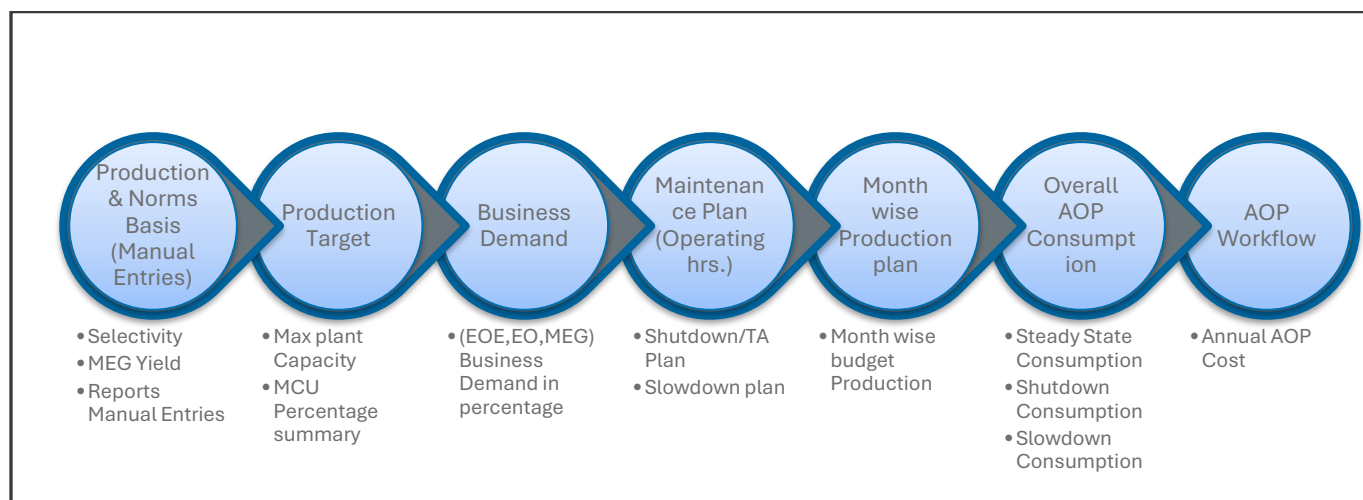


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Month wise Production plan

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production & Norms Basis

The production & norms basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data, including Yield, Selectivity, and Consumption, are configured under the production & norms basis screen to support Production and Overall AOP Consumption calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

The user must insert the TST value month wise and save it with a valid reason. This will provide the Latent heat against TST.

NOTE- Latent heat will be calculated only if the user provides the TST temperature; otherwise, the latent heat value will be set to 0.

In production & norms basis screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Report manual entry tab is provided to enter report specific manual entries for last budgeted FY actual values (e.g. Budgeted selectivity, Actual Selectivity, guaranteed selectivity and actual operating Hours)
- Option to import and export manually entered from production & norms basis screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.

Production & Norms Basis

Year:

2025-26

Vertical:

MEG

Site:

VMD

Plant:

EOEG

1

2

3

4

5

6

7

8

9

10

11

12

Production & Norms Basis

Shutdown/TA Activities

Slowdown Activities

Production Target

Business Demand

Maintenance Details

Month wise Production plan

Steady State Consumption (Norm/Quantity)

Shutdown Consumption (Quantity)

Slowdown Consumption (Norm/Quantity)

Overall AOP Consumption (Norm/Quantity)

Feed Stock Availability

MEG / VMD / EOEG

AOP Historical Period Basis

Start Date

01-04-2020

End Date

31-03-2025

Load

(Last refreshed data on: 20-05-2025 02:30 PM for the period from 01-04-2020 to 31-03-2025)

AOP Design Basis

1. AOP design basis is considered for the period from 01-04-2020 to 31-03-2025.

Configuration

Constants

Report Manual Entry (2024-25)

Export

Import

Save

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	As per AOP234
PEG Yield	%	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	As per AOP
Calculation input for Norm														
Predicted Selectivity	%	90.9	90.8	90.7	90.7	90.6	90.5	90.467	90.396	90.322	90.248	90.182	90.108	As per AOP
TST	C	264.5	265.7	266.8	268	269.1	270.1	271.1	272.1	273.111	274.396	275.482	276.568	update
Total HP Steam Consumption	TPH	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	AOP
Total MP Steam Consumption	TPH	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	AOP
Ethylene purity		0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	from dict update
Oxygen purity		0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	from dict update
Ethylene losses basis		1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	As per AOP
Oxygen losses basis		1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	from dict update
Cycle gas TT across Reactor	°C	42	42	42	42	42	42	42	42	42	42	42	42	From DCT

Configuration

Constants

Report Manual Entry (2024-25)

Export

Import

Save

Particulars	UOM	Value	Remark
▼ Constant			
Argon purge flow	kg/h		30.0 From DCT
Argon purge molecular weight			21.5 From DCT
BFW Feed Temperature	°C		160.0 From DCT
CO2 Molecular weight			44.01 From DCT
Condensate temperature in Oil Cooler	°C		216.0 From DCT
Cycle gas CP	kcal/kg/°C		0.4 From DCT
Cycle gas flow	kg/h		65000.0 From DCT
Decimals = unaccounted losses			1.02 From DCT
EO molecular weight			44.05 From DCT
Ethylene concentration in purge	mol%		35.0 From DCT
Ethylene content in CO2	fr		0.003 From DCT
Ethylene Molecular Weight			28.05 From DCTFrom DCT
Latent heat of steam from Oil cooler	kcal/kg		508.0 From DCT
Oxygen concentration in purge	mol%		5.8 From DCT
Oxygen molecular weight			32.0 From DCT
Water Cp	kcal/kg/°C		1.0 From DCT
CO2 Stoichiometric vs Actual Ratio			1.0019 From DCT
Ethylene loss via COD			5.09 From DCT
Phenomen loss via PPH			4.3 From DCT
DMW vs Steam Ratio r Process			1.83 From DCT
DMW vs Steam Ratio			1.02 From DCT

Configuration Constants Report Manual Entry (2024-25) Export Import Save														
Particulars	UOM	Apr-24	May-24	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Remark
Report Manual Entry														
Budgeted Selectivity	%	88.6	88.4	88.2	88	87.8	87.6	87.4	87.2	87	86.8	86.6	86.4	TEST
Actual Selectivity	%	89.22	89	88.77	88.68	88.49	88.34	88.16	87.93	87	86.8	86.6	86.4	TEST
Guaranteed Selectivity	%	86.2	85.9	85.6	85.3	85	84.7	84.3	84	83.9	83.7	83.4	83.1	TEST
Actual Operating Hours	Hrs	720	744	720	744	744	720	744	720	744	744	672	744	test3

Figure 2 Digitized AOP screen for Production & Norms Basis

2.1.1.2 Production Target

The Max Capacity (MC) of EOEG plants is defined on an EOE basis.

MC values for Ethylene Oxide (EO), Ethylene Oxide Equivalent (EOE) and Monoethylene Glycol (EOEG) will be captured separately.

The following logic has been formulated to derive month-wise EOE and EO MC values.

- Raw data from SAP (process order) is considered as basis to derive the MCU.
- Month wise average production is calculated for each year.
- Post filtering of outliers, top 10% datapoints are taken from the Month wise calculated average data sets.
- Production Target for EOE is the average of 10% data points.

The MEG MC value is determined using the following formula.

$$MEG\ MC = MEG\ Yield * (EOE\ MC - EO\ MC)$$

Where EOEG yield is a month-wise manual entry captured in the AOP digitization portal in the Production & norms basis screen, as seen in the screenshot below.

Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
▼ Yields														
MEG Yield	%	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367 / As per AOP234
PEG Yield	%	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072 / As per AOP

Figure 3 Digitized AOP screen for EOEG Yield

If the user needs to modify the production rate this can be done by providing a specific reason.

The AOP digitization portal captures a month-wise percentage summary, comparing each month's value against the highest MU value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.

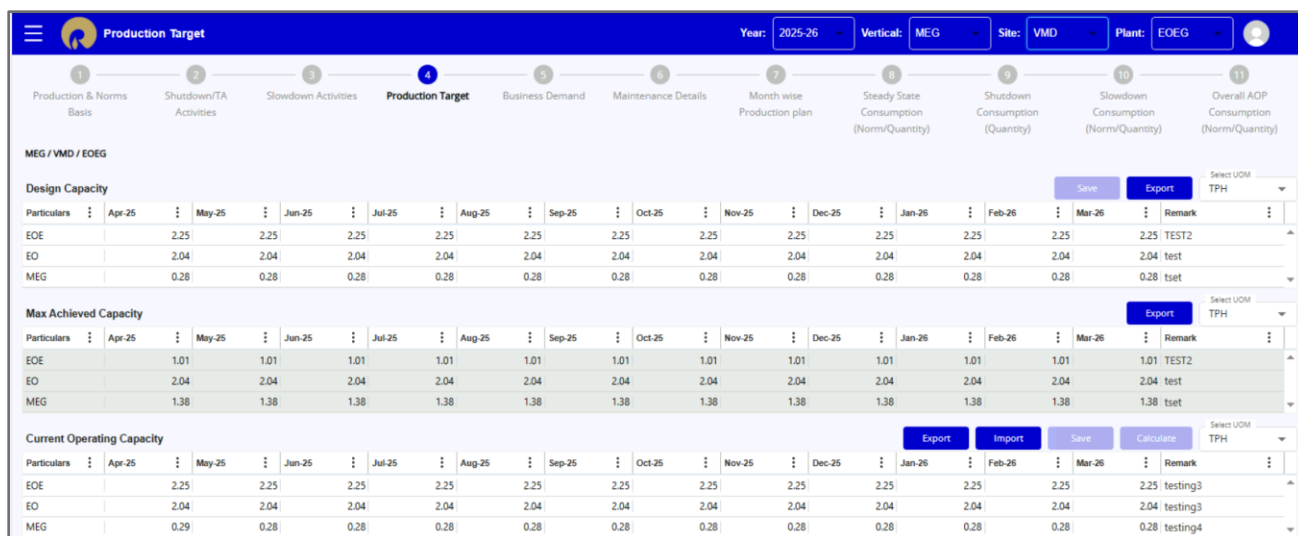


Figure 4 Digitized AOP screen for Production Target

The '**Calculate**' button on this screen allows the user to compute the EOEG MC value if any changes have been made to the EOEG Yield and Start and End date for AOP basis selection on the production & norms basis screen.

NOTE- AOP historical period basis added in production & norms basis screen where user can change Start date and End date to change the AOP design basis, accordingly MCU and Norms historical data will be considered for calculation.

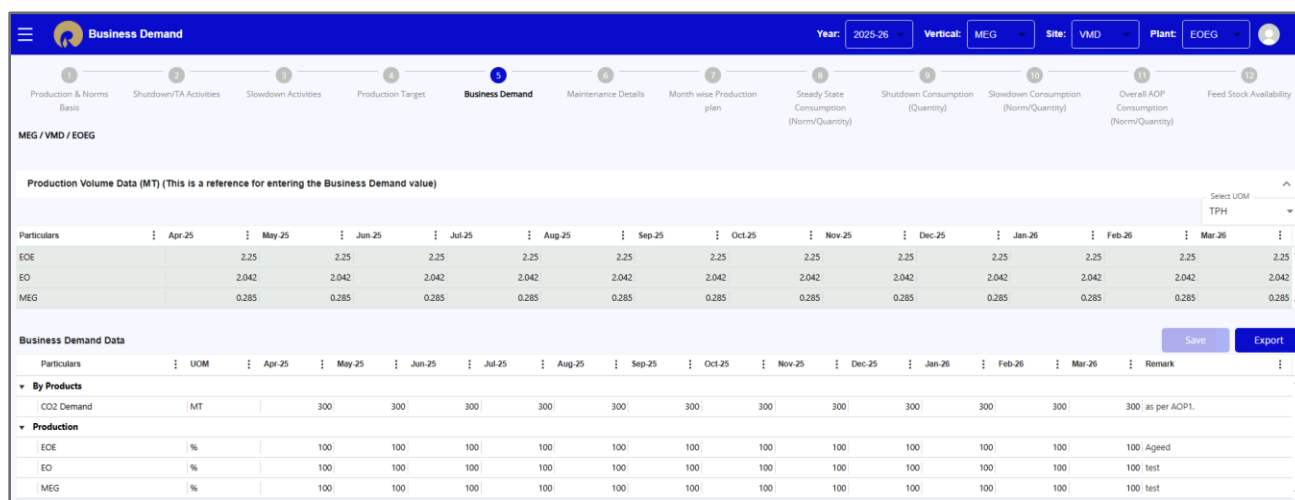
2.1.1.3 Business Demand

The production demand is considered as the maximum value from last year's achieved production values. The default value for EOE, EO and EOEG is 100%.

The business demand value can be modified by users upon providing a specific reason.

This information is captured in the AOP digitization portal as seen in the screenshot below.

NOTE- For the VMD EOEG plant, CO2 demand will be captured in this section.



Particulars	Apr 25	May 25	Jun 25	Jul 25	Aug 25	Sep 25	Oct 25	Nov 25	Dec 25	Jan 26	Feb 26	Mar 26
EOE	2.25	2.25	2.25	2.25	2.25	2.25	2.25	2.25	2.25	2.25	2.25	2.25
EO	2.042	2.042	2.042	2.042	2.042	2.042	2.042	2.042	2.042	2.042	2.042	2.042
MEG	0.285	0.285	0.285	0.285	0.285	0.285	0.285	0.285	0.285	0.285	0.285	0.285

Particulars	UOM	Apr 25	May 25	Jun 25	Jul 25	Aug 25	Sep 25	Oct 25	Nov 25	Dec 25	Jan 26	Feb 26	Mar 26	Remark
CO2 Demand	MT	300	300	300	300	300	300	300	300	300	300	300	300	as per AOP1.
EOE	%	100	100	100	100	100	100	100	100	100	100	100	100	Ageed
EO	%	100	100	100	100	100	100	100	100	100	100	100	100	test
MEG	%	100	100	100	100	100	100	100	100	100	100	100	100	test

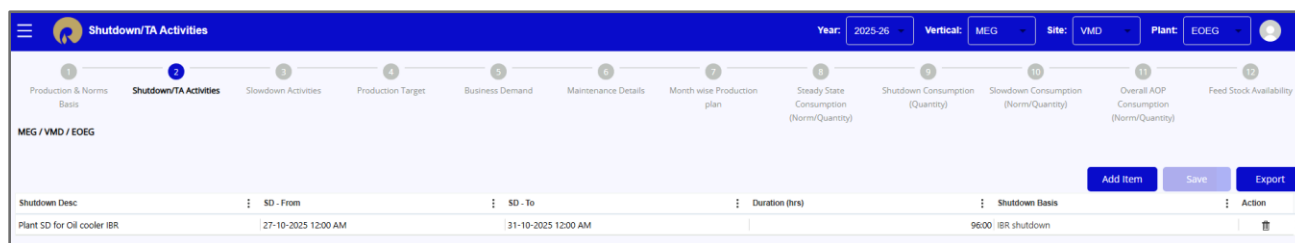
Figure 5 Digitized AOP screen for Business Demand

2.1.1.4 Maintenance/Shutdown/Slowdown/TA Activities

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The below screenshot depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.



Shutdown Desc	SD - From	SD - To	Duration (hrs)	Shutdown Basis	Action
Plant SD for Oil cooler IIR	27-10-2025 12:00 AM	31-10-2025 12:00 AM		9600 IIR shutdown	

Figure 6 Digitized AOP screen for Shutdown/Turnaround Activities

The Slowdown entries will have the inputs as shutdowns, along with the requirement of plant EOE/EO load (TPH basis) for the duration of the slowdown.

The slowdown activities can be calculated based on the slowdown rate and hours. This is called a Basis Shutdown, equivalent to Shutdown.

$$\text{Slowdown rate(TPH)} = (\text{MCU} - \text{Operating Production rate})$$

$$\text{Basis shutdown equivalent of Slowdown hrs} = (\text{Slowdown rate(TPH)} * \text{Slowdown hrs}) / \text{MCU}$$

Operating Production rate will be a manual entry in Slowdown activity screen and slowdown rate will

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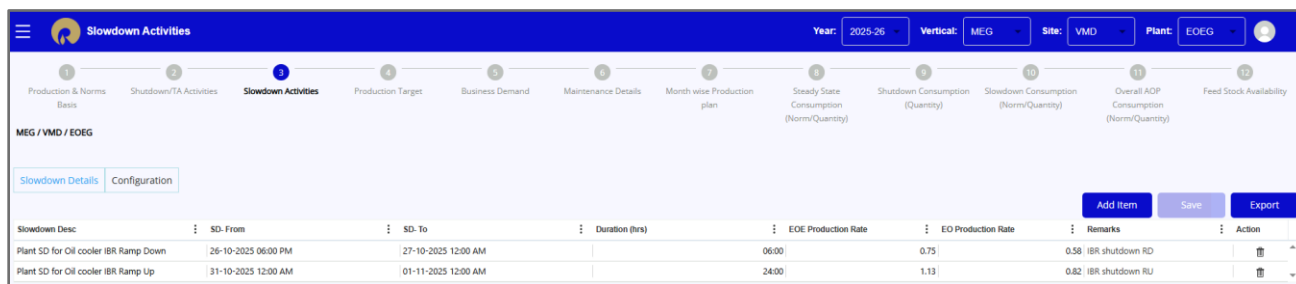
Document No.

September 18, 2025

be calculate at the backend as per above formula.

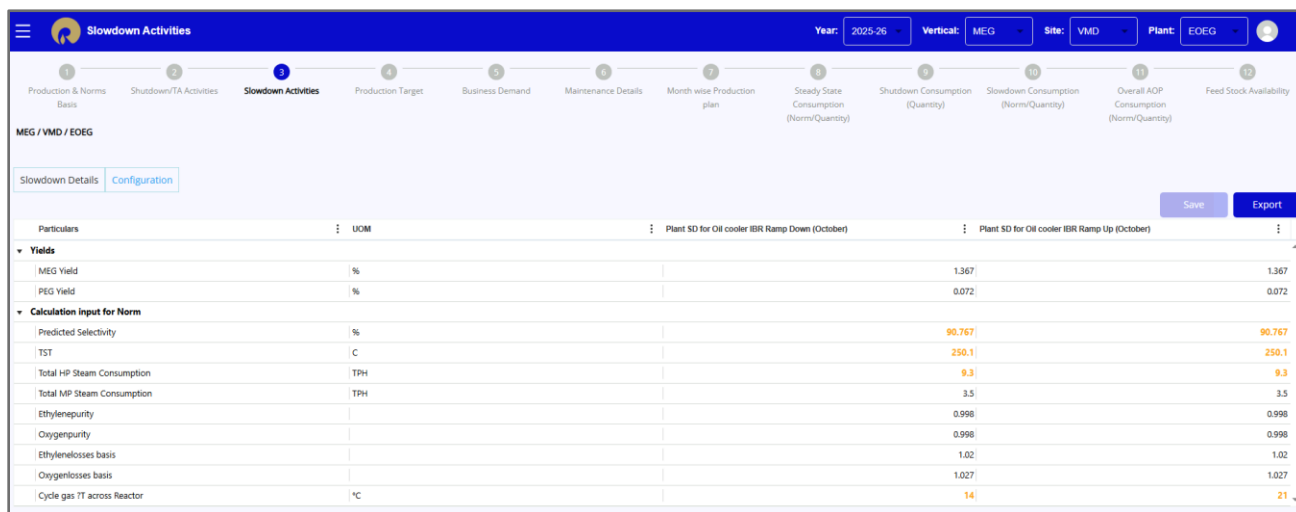
For each shutdown entry, two slowdown entries will be generated automatically, with the nomenclature as “#Shutdown Entry name. – Ramp down” and “#Shutdown Entry name. – Ramp Up”.

In Slowdown details tab, operating production rate-EOE (TPH) and operating production rate-EO (TPH) these two columns will be added in Slowdown activities screen as shown below.



Slowdown Desc	SD-From	SD-To	Duration (hrs)	EOE Production Rate	EO Production Rate	Remarks	Action
Plant SD for Oil cooler IBR Ramp Down	26-10-2025 06:00 PM	27-10-2025 12:00 AM		06:00	0.75	0.58 IBR shutdown RD	
Plant SD for Oil cooler IBR Ramp Up	31-10-2025 12:00 AM	01-11-2025 12:00 AM		24:00	1.13	0.82 IBR shutdown RU	

Separate tab (Configuration) will be provided as shown below in slowdown activity screen for the provision of manual entry related to production & norms basis screen parameters based on the type of slowdown. Number of rows mentioned in slowdown activity screen will be no of column for slow down manual entry screen.



Particulars	UOM	Plant SD for Oil cooler IBR Ramp Down (October)	Plant SD for Oil cooler IBR Ramp Up (October)
Yields			
MEG Yield	%	1.367	1.367
PEG Yield	%	0.072	0.072
Calculation input for Norm			
Predicted Selectivity	%	90.767	90.767
TST	C	250.1	250.1
Total HP Steam Consumption	TPH	9.3	9.3
Total MP Steam Consumption	TPH	3.5	3.5
Ethylene purity		0.998	0.998
Oxygen purity		0.998	0.998
Ethylene losses basis		1.02	1.02
Oxygen losses basis		1.027	1.027
Cycle gas TT across Reactor	°C	14	21

Figure 7 Digitized AOP screen for Slowdown Activities

The 'Add Item' button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

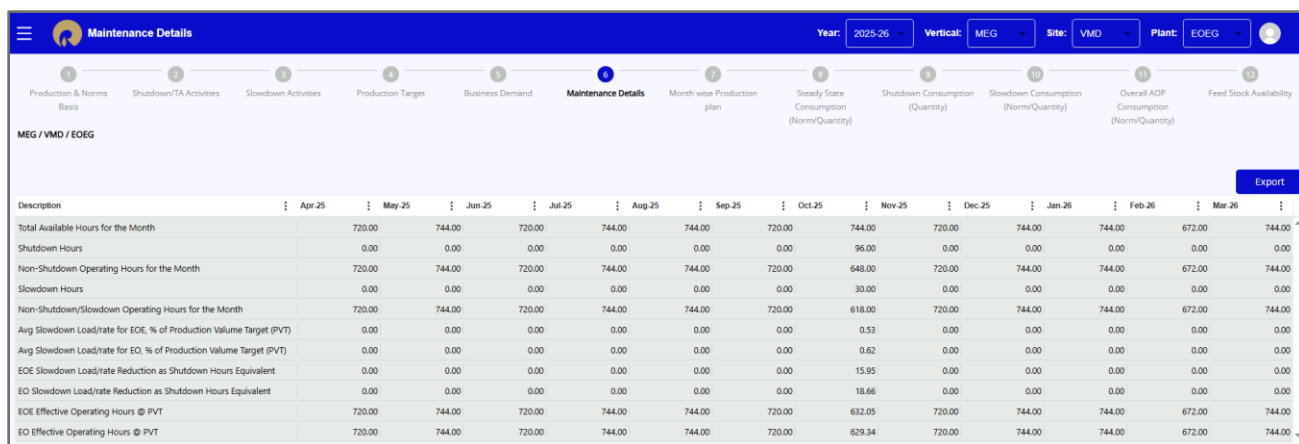
For each month, the following table is generated based on all inputs for shutdowns/slowdowns-

Description	April	...	March
Total Available Hours for the Month	$30 \times 24 = 720$		
Shutdowns Hours	<i>From the Routine Shutdown Plan Section</i>		
Non-Shutdown Operating Hours for the Month	<i>Total Available operating hours for the month – Shutdown hrs</i>		
Slowdown Hours	<i>From the Slowdown activities details</i>		
Non-Shutdown/Slowdown Operating Hours for the Month	Non Shutdown Operating Hours for the Month – Slowdown Hours		
Average Slowdown Load/rate for EOE, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i \times EOE \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EOE \text{ slowdown hrs}_i) \times EOE \text{ MC for Month}}$		
Average Slowdown Load/rate for EO, % of Production Volume Target (PVT)	$\frac{\sum_{i=1}^n (EO \text{ slowdown hrs}_i \times EO \text{ load/rate during slowdown}_i)}{\sum_{i=1}^n (EO \text{ slowdown hrs}_i) \times EO \text{ MC for Month}}$		
EOE Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EOE \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EOE \text{ slowdown hrs}_i)$		
EO Slowdown Load/rate Reduction as Shutdown Hours Equivalent	$EO \text{ Average slowdown load/rate, \% of PVT} \times \sum_{i=1}^n (EO \text{ slowdown hrs}_i)$		
EOE Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EOE Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		
EO Effective Operating Hours @ PVT	<i>Total Available operating hours for the month – Shutdown/TA hours – EO Slowdown load reduction as Shutdown hours Equivalent (other reasons) – EO Slowdown load reduction as Shutdown hours Equivalent (Rampup / RampDown)</i>		

Table 3 Maintenance Details

The screen below depicts the Maintenance Details Table, which is captured in the digitized platform.

NOTE- The below screen is read-only. The users cannot change the values.



Description	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	744.00
Shutdown Hours	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.00	0.00	0.00	0.00	0.00
Non-Shutdown Operating Hours for the Month	720.00	744.00	720.00	744.00	744.00	744.00	720.00	648.00	720.00	744.00	744.00	744.00
Slowdown Hours	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.00	0.00	0.00	0.00	0.00
Non-Shutdown/Slowdown Operating Hours for the Month	720.00	744.00	720.00	744.00	744.00	744.00	720.00	618.00	720.00	744.00	744.00	744.00
Avg Slowdown Load/Rate for EOE, % of Production Volume Target (PVT)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.53	0.00	0.00	0.00	0.00
Avg Slowdown Load/Rate for EO, % of Production Volume Target (PVT)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.62	0.00	0.00	0.00	0.00
EOE Slowdown Load/Rate Reduction as Shutdown Hours Equivalent	0.00	0.00	0.00	0.00	0.00	0.00	0.00	15.95	0.00	0.00	0.00	0.00
EO Slowdown Load/Rate Reduction as Shutdown Hours Equivalent	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.66	0.00	0.00	0.00	0.00
EOE Effective Operating Hours @ PVT	720.00	744.00	720.00	744.00	744.00	744.00	720.00	632.05	720.00	744.00	744.00	744.00
EO Effective Operating Hours @ PVT	720.00	744.00	720.00	744.00	744.00	744.00	720.00	629.34	720.00	744.00	744.00	744.00

Figure 8 AOP Maintenance Details Table

2.1.1.5 Calendar Year Based Production Calculations Details/Month wise Production plan

Using the Business Demand, Production Target and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\text{Production AOP (MT)} = (\text{Operating hours for EOE/EO in a month (hrs)} * \text{Percentage of product demand} * \text{MC Value of product(TPH)}) / 100$$

$$\text{Operating hours for MEG} = \text{Total Available hours of the month} - \text{Shutdown/TA hours of the month}$$

$$\text{Operating hours for EOE} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EOE Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

$$\text{Operating hours for EO} = \text{Total Available operating hours for the month} - \text{Shutdown/TA hours} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (other reasons)} - \text{EO Slowdown load reduction as Shutdown hours Equivalent (Ramp up/Ramp Down)}$$

Hence, month-wise production is,

$$\text{Total Budgeted Production} = \sum \text{All the values for the month}$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.

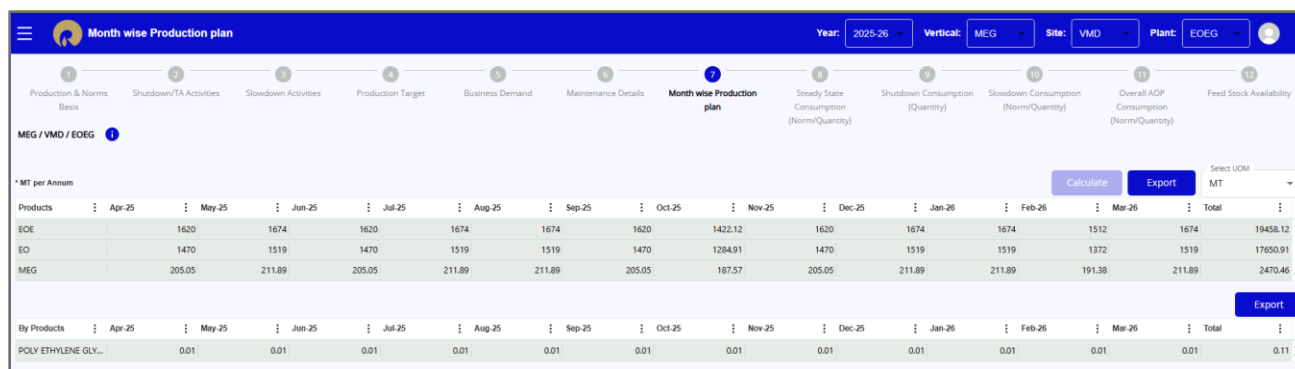


Figure 9 Digitized AOP screen for Month wise Production plan

The '**Calculate**' button on this screen allows users to recalculate Production values when changes are made to production & norms basis settings (e.g., EOEG yield), business demand, or when line items are added to slowdown and shutdown activities.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Consumption

All the specific Overall AOP Consumption are set with One-unit EOE as the basis.

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

1. Historical Best
2. Based on Calculations
3. Manual Entries

3.1.2.1.1 Historical Best

Below is the logic formulated to derive the Historical best Normal Operation Norms –

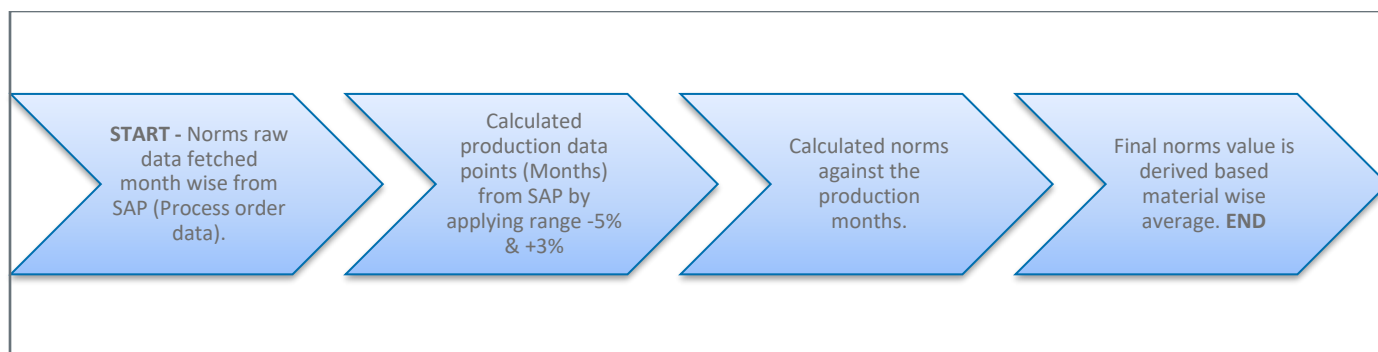


Figure 10 Digitized AOP Steps to derive month wise Historical Best Norms

- Norms raw data fetched month wise from SAP (Process order data).
- Production raw data fetched month wise from SAP (Process order data).
- Calculated production data points (Months) by applying range -5% & +3%
- Calculated norms against the production months
- Final norms value is derived based material wise average

The user has provision to overwrite/update the value by providing a valid reason for the historical best particulars.

The list of particulars to be derived using the Historical Best logic is specified in the DCT provided by the user. These particulars may vary from plant to plant.

3.1.2.1.2 Based on Calculations

For the EOEG vertical, particulars like ethylene, oxygen, HP Steam, BFW water etc. are formulated based on the stoichiometric calculations provided by the user in the DCT.

The formula can vary from plant to plant.

E.g. the Raw Material norms for ethylene and oxygen for VMD EOEG are derived using the calculations mentioned in the below table.

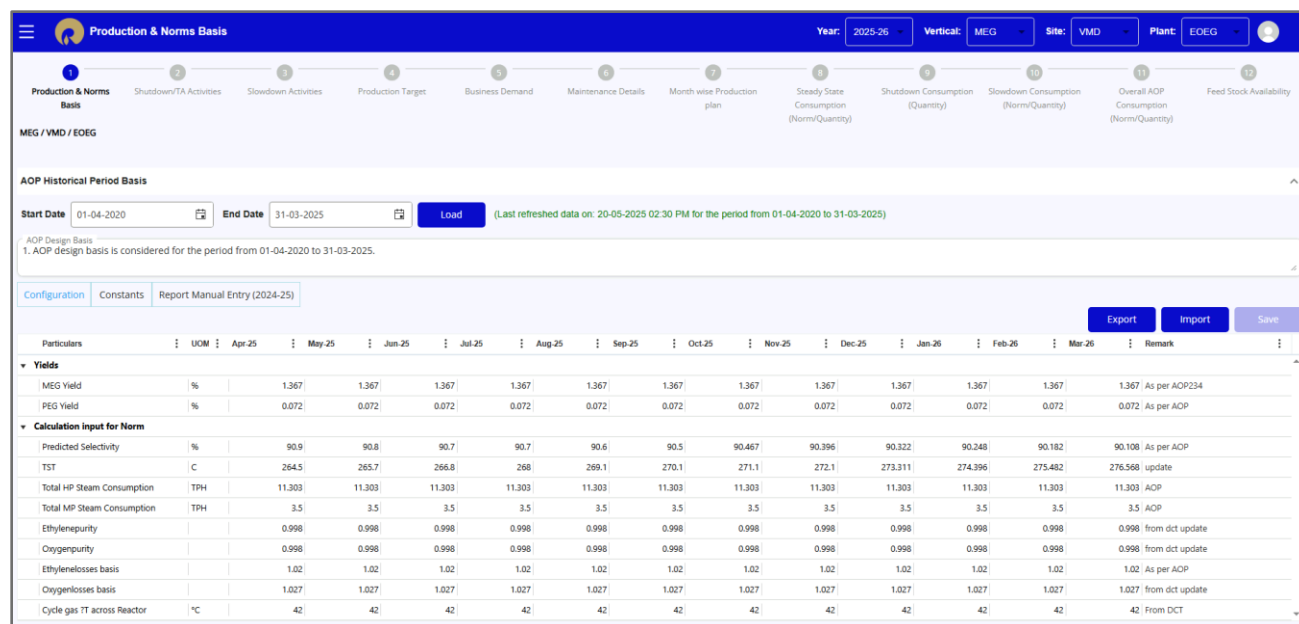
Plant	Category	Particular	Calculation
VMD EOEG	Raw Material	Ethylene	$\left(\frac{1.001 \times 28 \times 100}{44 \times \{selectivity\}} + \frac{[METHANE] \times 28 \times 0.28}{16 \times 0.54} \right) / 0.999 + (0.135 / 24 \times ([EOE] / [EOE \text{ effective Operating hrs}]))$ Where, 1.001 - decimals = EOE Losses 28 - Ethylene Molecular Weight 44 - EO molecular weight 0.28 - Ethylene concentration in argon purge (Mol %) 16 - Methane molecular weight 0.54 - Methane concentration in argon purge (Mol %) 0.999 - Ethylene Purity (wt%)

Plant	Category	Particular	Calculation
			0.135 - 1.9 - Allocated Ethylene consumptions (TPD)
		Oxygen	$(1.001 * (100 * 2.1818 / \{selectivity\} - 1.818) + ([METHANE] * 32 * 0.05 / 16 / 0.54)) / 0.997$ Where, 1.001 - decimals = EOE Losses 2.1818, 1.881 - Stoichiometric Ratios 32 - Oxygen molecular weight 0.05 - Oxygen concentration in argon purge (Mol %) 16 - Methane molecular weight 0.54 - Methane concentration in argon purge (Mol %) 0.997 - Oxygen Purity (wt%)

Table 4 Raw Material Norms calculations

The values for *losses and purity* can be captured from the Lab analysis or Past/last year's operations. The user has the provision to enter the value manually in the Production & norms basis screen.

Selectivity, Losses and Purity can be modified/added by the user in the Production & norms basis Screen under the Calculation Input for Norms Subsection.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Yields														
MEG Yield	%	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367	1.367 As per AOP234
PEG Yield	%	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072	0.072 As per AOP
Calculation input for Norm														
Predicted Selectivity	%	90.9	90.8	90.7	90.7	90.6	90.5	90.467	90.396	90.322	90.248	90.182	90.108	90.108 As per AOP
TST	C	264.5	265.7	266.8	268	269.1	270.1	271.1	272.1	273.111	274.196	275.482	276.568	276.568 update
Total HP Steam Consumption	TPH	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303	11.303 ACP
Total MP Steam Consumption	TPH	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5 ACP
Ethylene purity		0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998 from dict update
Oxygen purity		0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998 from dict update
Ethylene losses basis		1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02 As per ACP
Oxygen losses basis		1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027	1.027 from dict update
Cycle gas TT across Reactor	°C	42	42	42	42	42	42	42	42	42	42	42	42	42 From DCT

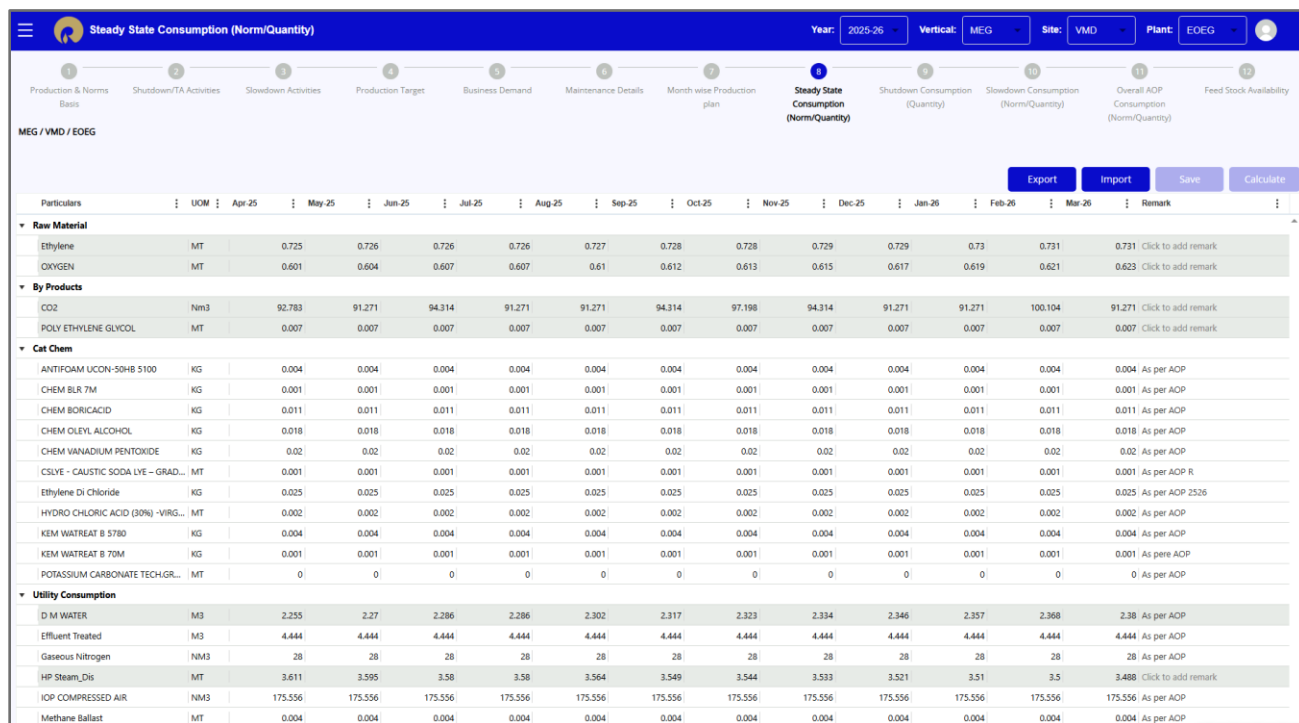
Figure 11 Digitized AOP screen for Catalyst Selectivity month wise

In the Steady State Consumption screen the cells for particulars with formula, e.g. HP Steam Dis, will be read only. If the values for such particulars are to be modified then the respective inputs should be modified, e.g. Total HP Steam consumption in the Production & norms basis Screen.

3.1.2.1.3 Manual Entries

The particulars under the Chemicals and Catalyst sections are manually entered by the user, who must also provide the corresponding norm values in the Normal Operation Norms screen.

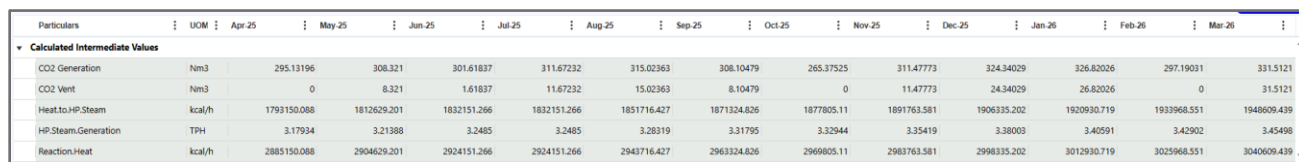
The screenshot below highlights the details captured in the digitized AOP platform in the Steady State Consumption screen.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Raw Material														
Ethylene	MT	0.725	0.726	0.726	0.726	0.727	0.728	0.728	0.729	0.729	0.73	0.731	0.731	Click to add remark
OXYGEN	MT	0.601	0.604	0.607	0.607	0.61	0.612	0.613	0.615	0.617	0.619	0.621	0.623	Click to add remark
By Products														
CO2	Nm3	92.783	91.271	94.314	91.271	91.271	94.314	97.198	94.314	91.271	91.271	100.104	91.271	Click to add remark
POLY ETHYLENE GLYCOL	MT	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	Click to add remark
Cat Chem														
ANTIFOAM UCON-50HB 5100	KG	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	As per AOP
CHEM BLR 7M	KG	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	As per AOP
CHEM BORICACID	KG	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011	As per AOP
CHEM CLEYL ALCOHOL	KG	0.018	0.018	0.018	0.018	0.018	0.018	0.018	0.018	0.018	0.018	0.018	0.018	As per AOP
CHEM VANADIUM PENTOXIDE	KG	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	As per AOP
CSLYE - CAUSTIC SODA LYE - GRAD...	MT	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	As per AOP R
Ethylene Di Chloride	KG	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025	As per AOP 2526
HYDRO CHLORIC ACID (30%) - VIRG...	MT	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	As per AOP
KEM WATREAT B 5780	KG	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	As per AOP
KEM WATREAT B 70M	KG	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	As per AOP
POTASSIUM CARBONATE TECH.GR...	MT	0	0	0	0	0	0	0	0	0	0	0	0	As per AOP
Utility Consumption														
D M WATER	M3	2.255	2.27	2.286	2.286	2.302	2.317	2.323	2.334	2.346	2.357	2.368	2.38	As per AOP
Effluent Treated	M3	4.444	4.444	4.444	4.444	4.444	4.444	4.444	4.444	4.444	4.444	4.444	4.444	As per AOP
Gaseous Nitrogen	NM3	28	28	28	28	28	28	28	28	28	28	28	28	As per AOP
HP Steam_Dis	MT	3.611	3.595	3.58	3.58	3.564	3.549	3.544	3.533	3.521	3.51	3.5	3.488	Click to add remark
IOP COMPRESSED AIR	NM3	175.556	175.556	175.556	175.556	175.556	175.556	175.556	175.556	175.556	175.556	175.556	175.556	As per AOP
Methane Ballast	MT	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	As per AOP

Figure 12 Digitized AOP screen for Steady State Consumption

The Normal Operation Normas screen also captures the Intermediate calculations required for Norms calculation as shown in the below screen shot –



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Calculated Intermediate Values													
CO2 Generation	Nm3	295.13196	308.321	301.61837	311.67232	315.02363	308.10479	265.37525	311.47773	324.34029	326.82026	297.19031	331.5121
CO2 Vent	Nm3	0	8.321	1.61837	11.67232	15.02363	8.10479	0	11.47773	24.34029	26.82026	0	31.5121
Heat.to.HP.Steam	kcal/h	1793150.088	1812629.201	1832151.266	1832151.266	1851716.427	1871324.826	1877805.11	1891763.581	1906335.202	1920930.719	1933968.551	1948609.439
HP.Steam.Generation	TPH	3.17934	3.21388	3.2485	3.2485	3.28319	3.31795	3.32944	3.35419	3.38003	3.40591	3.42902	3.45498
Reaction.Heat	kcal/h	2885150.088	2904629.201	2924151.266	2924151.266	2943716.427	2963324.826	2969805.11	2983763.581	2998335.202	3012930.719	3025968.551	3040609.439

Figure 14 Digitized AOP screen for Steady State Consumption (Calculated Intermediate values)

The **'Calculate'** button on the screen will enable if the user makes any change in the dependent inputs in the Production & norms basis screen.

The **'Save'** button on the above screen enables the user to save any norm value updated manually for historical best norms for cat chem.

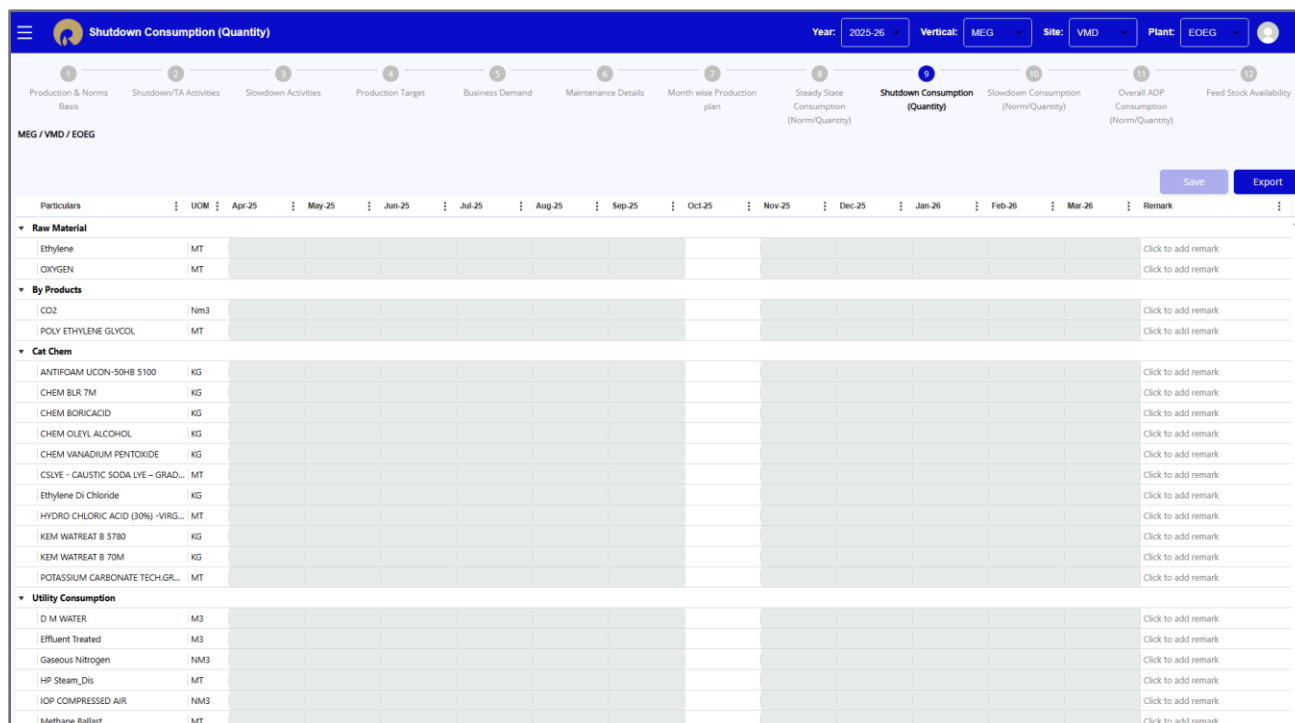
2.1.2.2 Shutdown Consumption

The Shutdown Consumption are recorded for the months in which shutdown activities are planned, as specified in the Shutdown Activities screen.

The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc..

The user must enter the consumption data (absolute values) for each line item.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
MEG / VMD / EOEG														
Raw Material														
Ethylene	MT													Click to add remark
OXYGEN	MT													Click to add remark
By Products														
CO2	Nm3													Click to add remark
POLY ETHYLENE GLYCOL	MT													Click to add remark
Cat Chem														
ANTIFOAM UCOR-50HB 5100	KG													Click to add remark
CHEM BLR 7M	KG													Click to add remark
CHEM BORICACID	KG													Click to add remark
CHEM OLEYL ALCOHOL	KG													Click to add remark
CHEM VANADIUM PENTOXIDE	KG													Click to add remark
CSLYE - CAUSTIC SODA LYE - GRAD...	MT													Click to add remark
Ethylene Di Chloride	KG													Click to add remark
HYDRO CHLORIC ACID (30%) -VIRG...	MT													Click to add remark
KEM WATREAT B 5780	KG													Click to add remark
KEM WATREAT B 70M	KG													Click to add remark
POTASSIUM CARBONATE TECH.GR...	MT													Click to add remark
Utility Consumption														
D M WATER	M3													Click to add remark
Effluent Treated	M3													Click to add remark
Gaseous Nitrogen	NM3													Click to add remark
HP Steam_Dis	MT													Click to add remark
IOP COMPRESSED AIR	NM3													Click to add remark
Methane Ballast	MT													Click to add remark

Figure 13 Digitized AOP screen for Shutdown Consumption

2.1.2.3 Slowdown Consumption

Slowdown Consumption are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The values for Slowdown Consumption are same as the Steady State Consumption for the month.

NOTE- Slowdown wise norms calculation will be provided by RIL.

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

Slowdown Consumption (Norm/Quantity)

Year: 2025-26 Vertical: MEG Site: VMD Plant: EDEG

1

Production & Norms Basis

2

Shutdown/TA Activities

3

Slowdown Activities

4

Production Target

5

Business Demand

6

Maintenance Details

7

Month wise Production plan

8

Steady State Consumption (Norm/Quantity)

9

Shutdown Consumption (Quantity)

10

Slowdown Consumption (Norm/Quantity)

11

Overall AOP Consumption (Norm/Quantity)

12

Feed Stock Availability

MEG / VMD / EDEG

Save

Calculate

Export

Particulars	UOM	Plant SD for Oil cooler IBR Ramp Down (October)	Plant SD for Oil cooler IBR Ramp Up (October)
▼ Raw Material			
Ethylene	MT		0.743
OXYGEN	MT		0.611
▼ By Products			
CO2	Nm3		94.139
POLY ETHYLENE GLYCOL	MT		0.016
▼ Cat Chem			
ANTIFOAM UCON-50HB 5100	KG		
CHEM BLR 7M	KG		
CHEM BORICACID	KG		
CHEM OLEYL ALCOHOL	KG		
CHEM VANADIUM PENTOXIDE	KG		
CSLYE - CAUSTIC SODA LYE - GRAD...	MT		
Ethylene Di Chloride	KG		
HYDRO CHLORIC ACID (30%) -VIRG...	MT		
KEM WATREAT B 5780	KG		
KEM WATREAT B 70M	KG		
POTASSIUM CARBONATE TECH.GR...	MT		
▼ Utility Consumption			
D M WATER	M3		3.902
Effluent Treated	M3		13.333
Gaseous Nitrogen	NM3		84
HP Steam_Dis	MT		10.967
IOP COMPRESSED AIR	NM3		526.667
Methane Ballast	MT		0.012

Figure 14 Digitized AOP screen for Slowdown Consumption

2.1.2.4 Budget Norms/Overall AOP Consumption

Using the values of Steady State Consumption, Shutdown Consumptions and Slowdown Consumption, the final Overall AOP Consumption value is calculated as below –

Consumption AOP for months without shutdown or slowdown = Normal Operations Norms

*Consumption AOP for months with shutdown or slowdown = Total Consumption/
Total production*

Where,

*Total Consumption = (Normal Operation Norms * EOE MC value * EOE operating hours) + $\sum_{i=1}^n$ (Slowdown Norms * EOE slowdown hours * EOE slowdown rate)_n + Shutdown Consumption*

*Total Production = (EOE MC value * EOE operating hours) + $\sum_{i=1}^n$ (EOE slowdown hours * EOE slowdown rate)_n*

The below screenshot depicts the Shutdown Consumption captured in the AOP digitized platform.

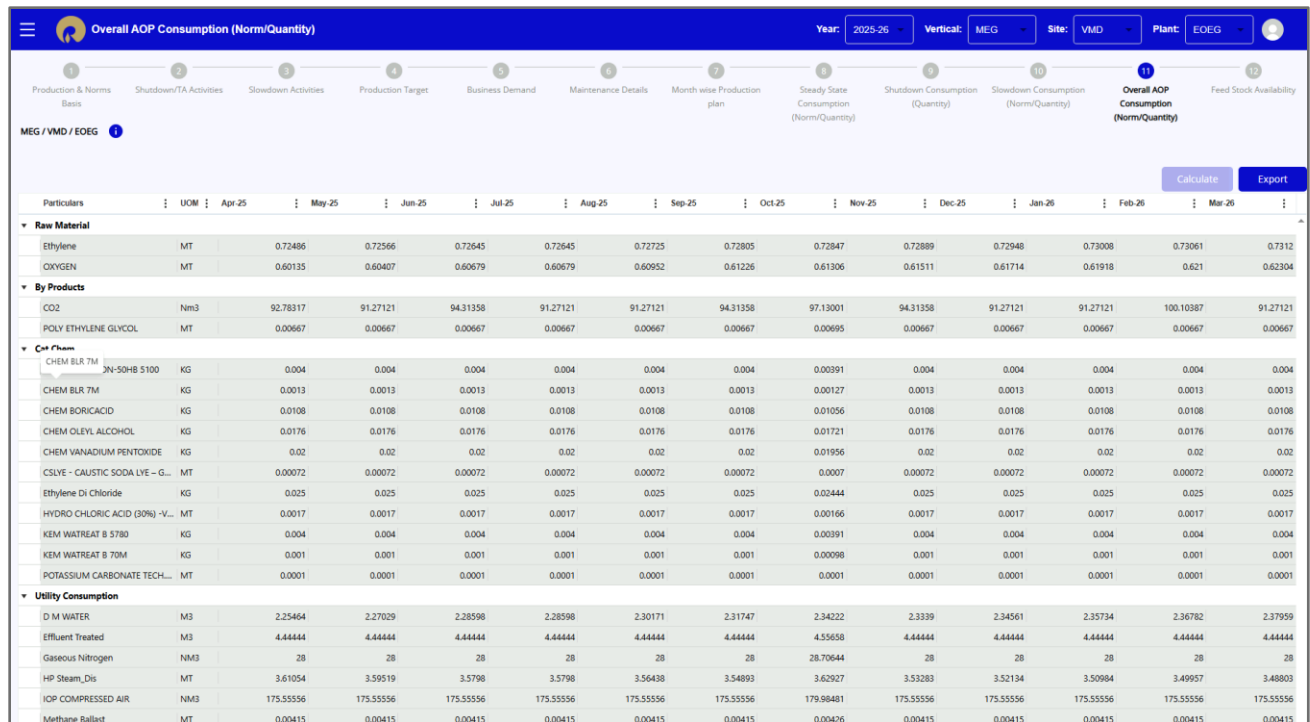
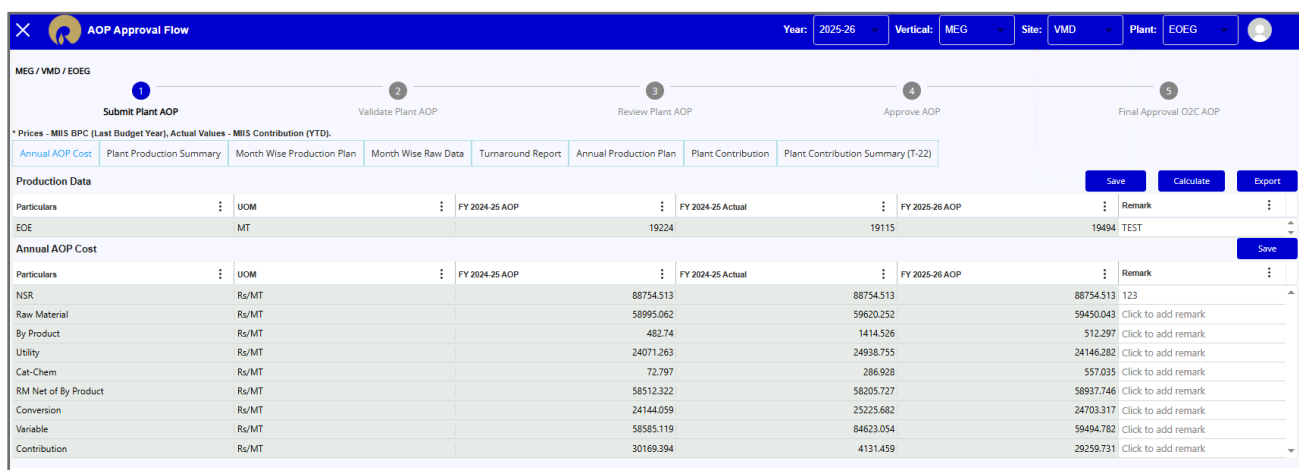


Figure 15 Digitized AOP screen for Budget Norms or Overall AOP Consumption

3. AOP Reports

- Annual AOP Cost.
- Plant Production Summary (T-14).
- Month wise Production Plan (T – 16)
- Month wise Raw data (T – 18).
- Turnaround Report (T – 19A).
- Annual Production Plan (T – 15).
- Plant Contribution (T – 21).
- Plant Contribution Summary (T-22)

The below screenshot depicts the AOP report captured in the AOP digitized platform.



AOP Approval Flow | Year: 2025-26 | Vertical: MEG | Site: VMD | Plant: EOEg

MEG / VMD / EOEg

1 Submit Plant AOP | 2 Validate Plant AOP | 3 Review Plant AOP | 4 Approve AOP | 5 Final Approval Q2C AOP

* Prices - MII BPC (Last Budget Year), Actual Values - MII Contribution (YTD).

Annual AOP Cost | Plant Production Summary | Month Wise Production Plan | Month Wise Raw Data | Turnaround Report | Annual Production Plan | Plant Contribution | Plant Contribution Summary (T-22)

Production Data

Particulars	UOM	FY 2024-25 AOP	FY 2024-25 Actual	FY 2025-26 AOP	Remark
EOE	MT		19224	19115	19494 TEST

Annual AOP Cost

Particulars	UOM	FY 2024-25 AOP	FY 2024-25 Actual	FY 2025-26 AOP	Remark
NSR	Rs/MT		88754.513	88754.513	88754.513 123
Raw Material	Rs/MT		58995.062	59620.252	59450.043 Click to add remark
By Product	Rs/MT		482.74	1414.526	512.297 Click to add remark
Utility	Rs/MT		24071.263	24938.755	24146.282 Click to add remark
Cat-Chem	Rs/MT		72.797	286.928	557.035 Click to add remark
RM Net of By Product	Rs/MT		58912.322	58205.727	58937.746 Click to add remark
Conversion	Rs/MT		24144.059	25225.682	24703.317 Click to add remark
Variable	Rs/MT		58585.119	84623.054	59494.782 Click to add remark
Contribution	Rs/MT		30169.394	4131.459	29259.731 Click to add remark

Figure 16 Digitized AOP screen for Annual AOP Cost

AOP Approval Flow

Year: 2025-26

Vertical: MEG

Site: VMD

Plant: EOEg

MEG / VMD / EOEg

1

2

3

4

5

Submit Plant AOP

Validate Plant AOP

Review Plant AOP

Approve AOP

Final Approval Q2C AOP

* Prices - MIIS BPC (Last Budget Year), Actual Values - MIIS Contribution (YTD).

Annual AOP Cost

Plant Production Summary

Month Wise Production Plan

Month Wise Raw Data

Turnaround Report

Annual Production Plan

Plant Contribution

Plant Contribution Summary (T-22)

Plant Production Summary (T-14)

Save

SL.No	Item	2024-25	2025-26	Variance wrt current year budget		Variance wrt current year actuals		Remarks	
		Budget	Actual	Budget	MT	%	MT	%	
1	EOE	MT	19223.94	19114.75	19494	270.06	1.4	379.25	1.98 Click to add remark
2	MEG	MT	2432.94	3518.85	2469.39	36.45	1.5	-1049.46	-29.92 Click to add remark
3	EO	MT	17444	16194.77	17689	245	1.4	1494.23	9.23 Click to add remark
4	Production	MT	19223.94	19114.75	19494	270.06	1.4	379.25	1.98 Click to add remark
5	Operating Hours	Hrs	8760	8400	8664	-96	-1.1	264	3.14 Click to add remark
6	ProductionRate	TPH	2.19	2.28	2.25	0.06	2.53	-0.03	-1.12 Click to add remark
7	NSR	Rs/MT	88754.51	88754.51	88754.51	0	0	0	0 Click to add remark
8	Raw Material	Rs/MT	58995.06	59620.25	59450.04	-454.98	-0.77	170.21	0.29 Click to add remark
9	By Product	Rs/MT	482.74	1414.53	512.3	29.56	6.12	-902.23	-63.78 Click to add remark
10	RM Net of By Product	Rs/MT	58512.32	58205.73	58937.75	-425.42	-0.73	-732.02	-1.26 Click to add remark
11	Cat-Chem	Rs/MT	72.8	286.93	557.04	-484.24	-665.19	-270.11	-94.14 Click to add remark
12	Utility	Rs/MT	24071.26	24938.75	24146.28	-75.02	-0.31	792.47	3.18 Click to add remark
13	Conversion	Rs/MT	24144.06	25225.68	24703.32	-559.26	-2.32	522.37	2.07 Click to add remark
14	Variable	Rs/MT	58585.12	84623.05	59494.78	-909.66	-1.55	25128.27	29.69 Click to add remark
15	Contribution	Rs/MT	30169.39	4131.46	29259.73	-909.66	-3.02	25128.27	608.22 Click to add remark
16	Contribution Rs Lacs	Rs lacs	5799.74	789.72	5703.89	-95.85	-1.65	4914.17	622.27 Click to add remark

Figure 17 Digitized AOP screen for Production Plant Summary

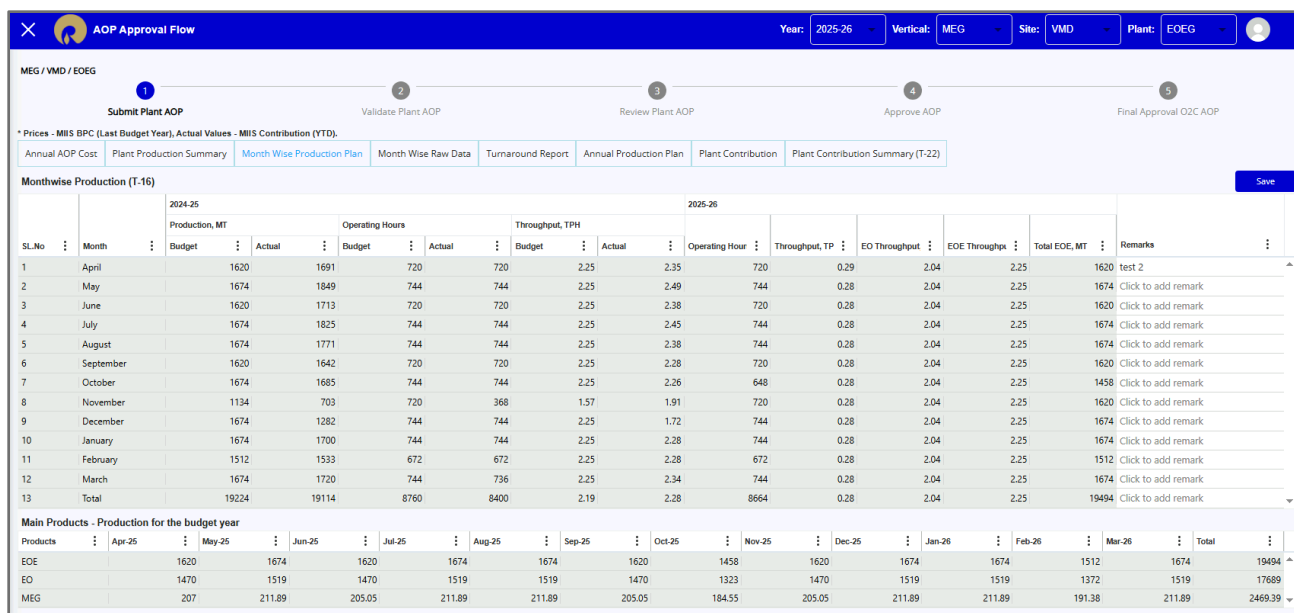


Figure 18 Digitized AOP screen for Month Wise Production Plan

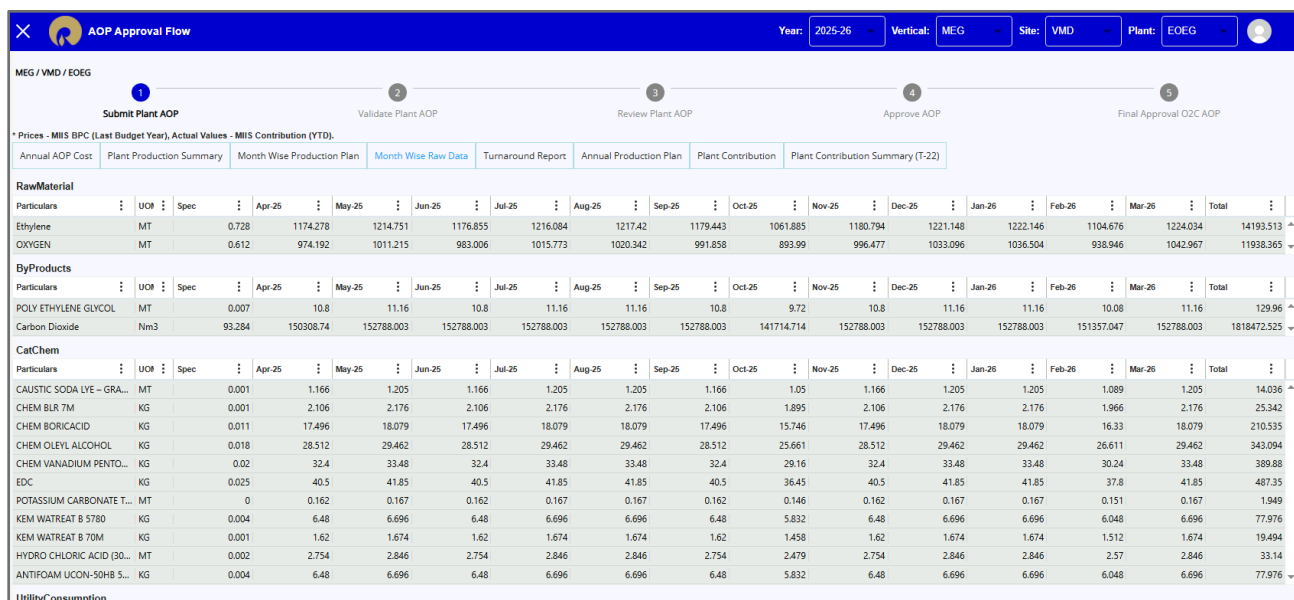


Figure 19 Digitized AOP screen for Month Wise Raw Data

AOP Approval Flow

Year: 2025-26 Vertical: MEG Site: VMD Plant: EOEG

MEG / VMD / EOEG

1 Submit Plant AOP

2 Validate Plant AOP

3 Review Plant AOP

4 Approve AOP

5 Final Approval Q2C AOP

* Prices - MIIIS BPC (Last Budget Year), Actual Values - MIIIS Contribution (YTD).

Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data **Turnaround Report** Annual Production Plan Plant Contribution Plant Contribution Summary (T-22)

Turnaround Details (T-19A)

SL.No	Activities	From	To	Duration, hrs	Remark
1	Plant SD for Oil cooler IBR	27-Oct-25	31-Oct-25	96.00	Click to add remark

Turnaround details for the previous years data since commissioning

SL.No	Activities	Turnaround Period From	Turnaround Period To	Duration, hrs	Period in Months	Remark	Action
1	Plant SD for Oil cooler IBR	27-Oct-25	31-Oct-25	96		Click to add remark	

Figure 20 Digitized AOP screen for Turnaround Report

AOP Approval Flow

Year: 2025-26 Vertical: MEG Site: VMD Plant: EOEG

MEG / VMD / EOEG

1 Submit Plant AOP

2 Validate Plant AOP

3 Review Plant AOP

4 Approve AOP

5 Final Approval Q2C AOP

* Prices - MIIIS BPC (Last Budget Year), Actual Values - MIIIS Contribution (YTD).

Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report **Annual Production Plan** Plant Contribution Plant Contribution Summary (T-22)

Plant Production Plan (T-15) - Assumptions & remarks

SL.No	Assumptions & remarks
1	1 MT EOE = 1 MT EO
2	1 MT EOE = 1.36808 MEG
2	1 MT EOE = 1.23552 MEG
2	1 MT EOE = 1.367 MEG

Max hourly rate achieved

SL.No	Max hourly rate achieved	Value	UOM
1	Recorded max daily production	112.63	TPD
2	Recorded date (DD/MM/YY)	28-Feb-25	
3	Equivalent max hourly production	4.69	TPH

Calculation of Operating hours

SL.No	Calculation of Operating hours	Value	Hours
1	Total available hours	8760	Hrs
2	Planned turnaround	96	Hrs
3	Planned shutdown other than turnaround	0	Hrs
4	Routine shutdowns	0	Hrs
5	Slowdown hours	0	Hrs
6	Total Shutdown Hours	96	Hrs
7	Plant Operating Hours	8664	Hrs

Figure 21 Digitized AOP screen for Annual Production Plan

AOP Approval Flow											
MEG / VMD / EOEG											
1 2 3 4 5 Submit Plant AOP Validate Plant AOP Review Plant AOP Approve AOP Final Approval Q2C AOP											
Prices - MIIIS BPC (Last Budget Year), Actual Values - MIIIS Contribution (YTD). Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report Annual Production Plan Plant Contribution Plant Contribution Summary (T-22)											
Plant Contribution (T-21) - MEG											
Product mix and Production											
SL.No	Product name	UOM	Rs/UOM	Production, MT		2024-25		2025-26		2025-26	
				Budget	Actual	Budget	Actual	Budget	Actual	Budget	Actual
1	EOE	MT			19223.94			19114.75		19494	
2	MEG	MT		48312	2432.94			3518.85		2469.39	
3	EO	MT		91072	17444			16194.77		17689	
4	NSR	Rs / MT			88754.51			86054.17		88759.86	
5	Production	MT			19223.94			19114.75		19494	
6	Value of production Rs Crs				170.62			164.49		173.03	
7	Variance % wrt current year Budget									1.4	
8	Variance % wrt current year Actual									5.2	
By products											
SL.No	By product name	UOM	Rs/UOM	Norm Unit/MT		2024-25		2025-26		2025-26	
				Budget	Actual	Budget	Actual	Budget	Actual	Budget	Actual
1	Carbon Dioxide	NM3	2	79.99	86.63			159.97		173.26	186.78
2	POLY ETHYLENE GLYCOL	MT	46621	0.01	0.03			312.36		1235.46	312.36
3	Total Rs / MT			472.33	1408.71			499.14		1408.71	499.14
4	ByProducts Value Rs Crs			0.91	2.69			0.97			
5	variance % wrt current year Budget							7.2			
6	variance % wrt current year Actual							-63.9			

Figure 22 Digitized AOP screen for Plant Contribution

AOP Approval Flow											
MEG / VMD / EOEG											
1 2 3 4 5 Submit Plant AOP Validate Plant AOP Review Plant AOP Approve AOP Final Approval Q2C AOP											
Prices - MIIIS BPC (Last Budget Year), Actual Values - MIIIS Contribution (YTD). Annual AOP Cost Plant Production Summary Month Wise Production Plan Month Wise Raw Data Turnaround Report Annual Production Plan Plant Contribution Plant Contribution Summary (T-22)											
Plant Contribution Summary (T-22)											
Product mix and Production											
S.no	Product name	UOM	Rs/UOM	Production, MT		2022-23		2023-24		2024-25	
				Actual	Budget	Actual	Budget	Actual	Budget	Actual	Budget
1	EO	MT	97589	15878.41	9985.68			16399.32		16194.77	17689
2	EOE	MT	95348	18358.85	11464.94			19032.17		19114.75	19494
3	MEG	MT	54148	3042.73	1838.51			3390.37		3518.85	2469.39
4	Total	Rs/MT		93378.18	93680.75			93734.71		92649.39	95412.13
5	Production	MT		18358.85	11464.94			19032.17		19114.75	19494
6	Value of production	Rs. Crs		171.43	107.4			178.4		177.1	186
By products											
S.no	Product Name	UOM	Rs/UOM	Specific Consumptions		2022-23		2023-24		2024-25	
				Actual	Budget	Actual	Budget	Actual	Budget	Actual	Budget
1	Carbon Dioxide	NM3	2	0.25	116	112.86	86.63	1818472.53	0.49	231.99	173.26
2	POLY ETHYLENE GLY...	MT	47681	0.02	0.02	0.01	0.03	23615.85390039162	999.06	869.58	1263.22
3	Total	Rs/MT		999.56	1101.57	906.69	1436.47	999.56	1101.57	906.69	1436.47
4	Byproduct Value Rs...	Rs. Crs		1.84	1.26	1.73	2.75	23615.85			
Raw material											
S.no	Product Name	UOM	Rs/UOM	Specific Consumptions		2022-23		2023-24		2024-25	
				Actual	Budget	Actual	Budget	Actual	Budget	Actual	Budget

Figure 23 Digitized AOP screen Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by VMD Site team